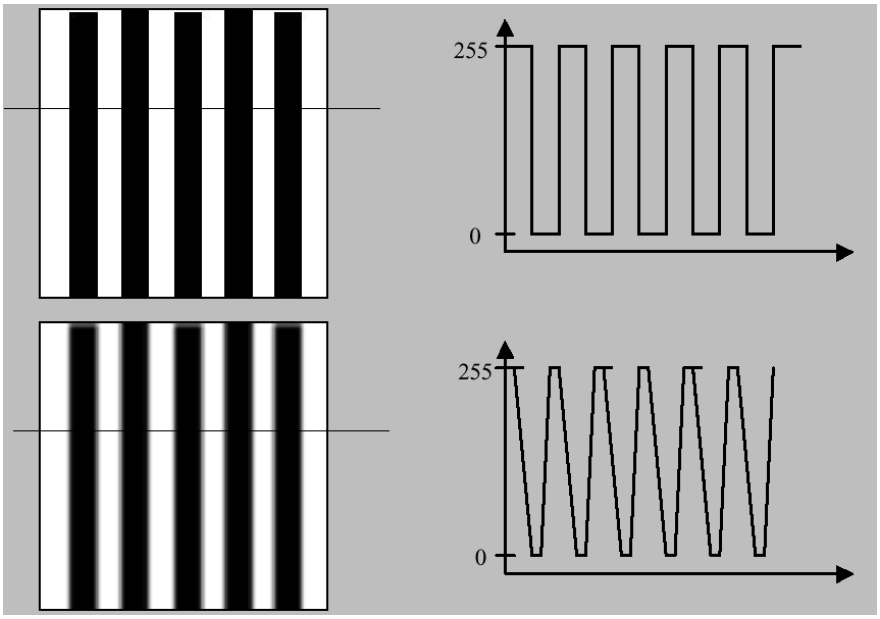
Computer Vision
Lecture 15: Filtering

Yeungnam University
MSP lab.

Ho-Youl JUNG

Smoothing (Blurring)



Smoothing (Blurring)

- Remove finer details of image
- Reduce noise, but lower the contrast
- Low-pass filtering
- Averaging filter
  - All coefficients are equal, simple averaging
  - 1/(nxn) in nxn mask (Local Averaging)
- Gaussian filter

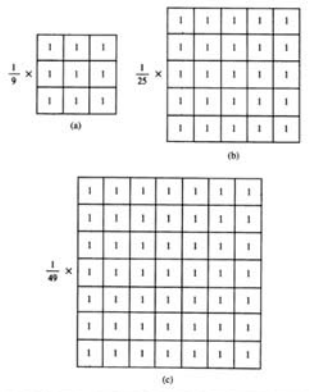


Figure 4.21 Spatial lowpass filters of various sizes.

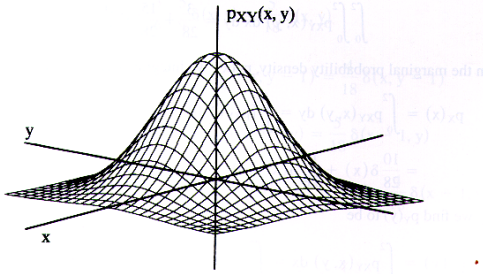


Figure 3.13 An example of a probability density function for the joint random variables (X, Y)

Smoothing (Blurring) : Averaging filter

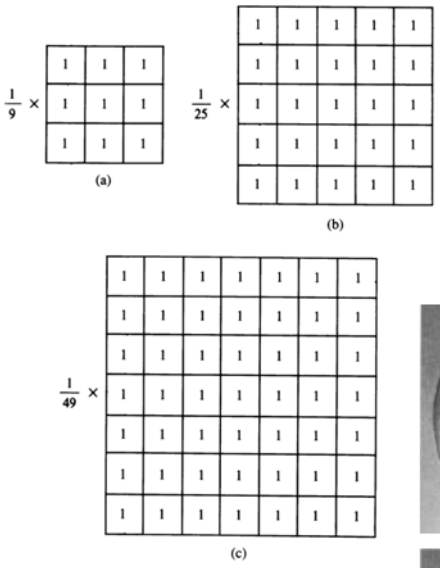
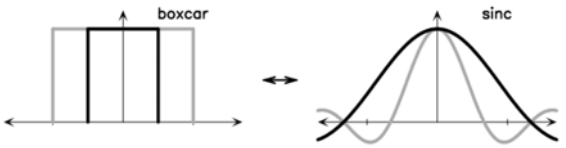
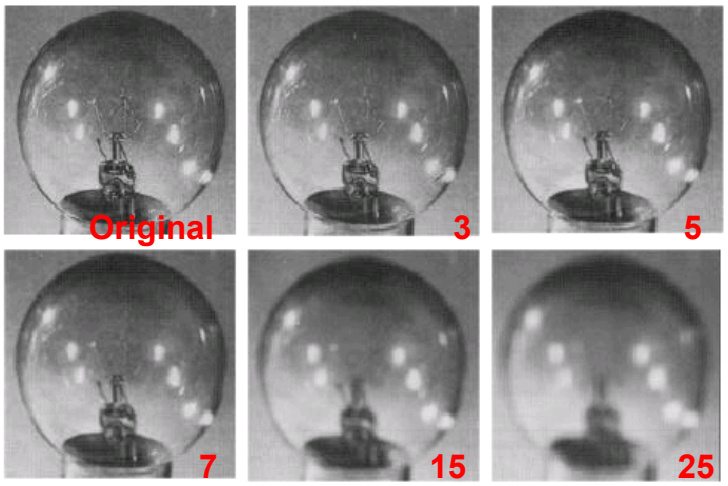


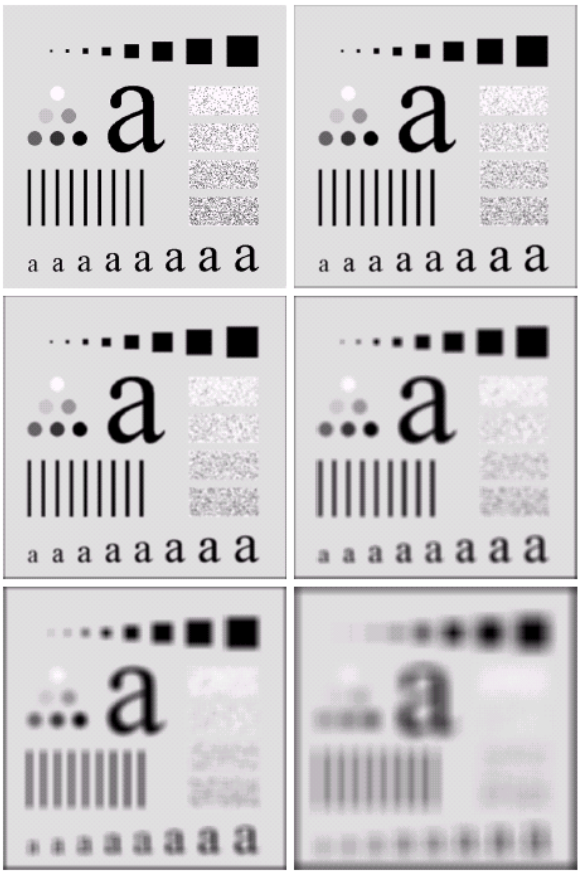
Figure 4.21 Spatial lowpass filters of various sizes.



https://www.cv.nrao.edu/~sransom/web/A1.html

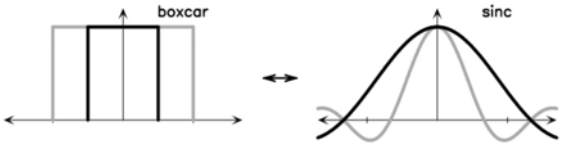


Smoothing (Blurring) : Averaging filter



a b
c d
e f

FIGURE 3.35 (a) Original image, of size 500 × 500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $n = 3, 5, 9, 15$, and 35, respectively. The black squares at the top are of sizes 3, 5, 9, 15, 25, 35, 45, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their gray levels range from 0% to 100% black in increments of 20%. The background of the image is 10% black. The noisy rectangles are of size 50 × 120 pixels.



https://www.cv.nrao.edu/~sransom/web/A1.html

Aliasing effect

Smoothing (Blurring)



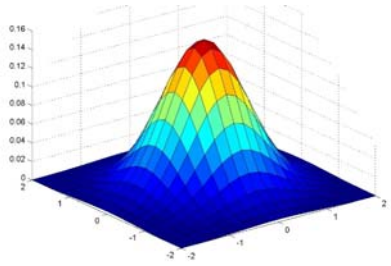


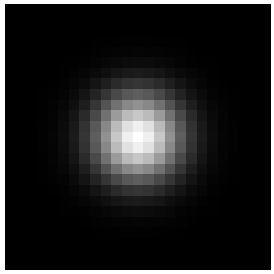
How to convolve color image?

- **convolve intensity component of HSI image**
- **convolve each channels of RGB image**

Smoothing (Blurring) : Gaussian Filter

$$G_\sigma = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$





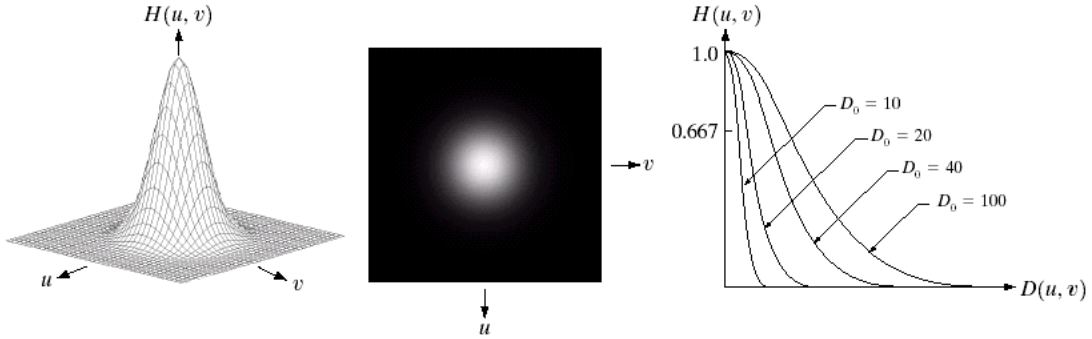
0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, σ = 1

Smoothing (Blurring) : Gaussian Filter

$$G_\sigma = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

$$Y(u,v) = F(u,v)\, G(u,v)$$

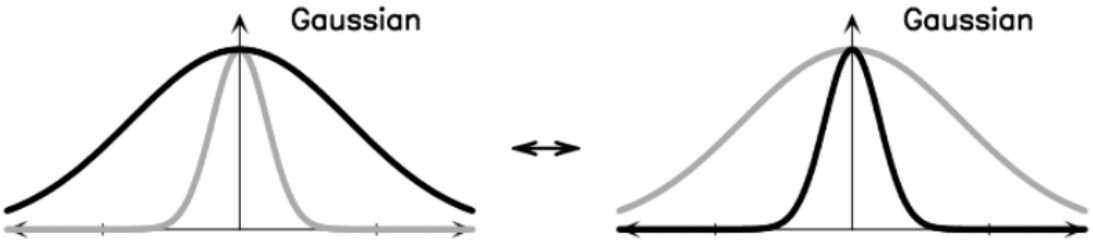


a b c

FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.

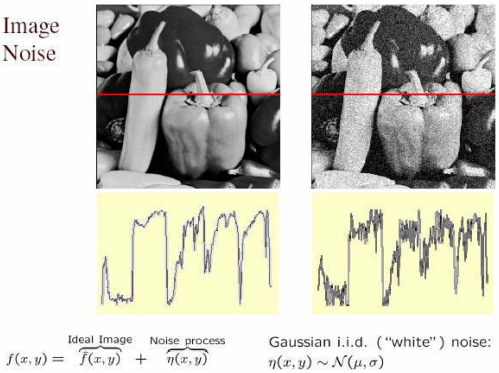
Fourier Transform of Gaussian Filter

$$g(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{t^2}{2\sigma^2}} \quad \overset{F.T.}{\Leftrightarrow} \quad G(\omega) = e^{-\frac{\omega^2\sigma^2}{2}}$$



Noise

- **Salt and pepper noise**: contains random occurrences of black and white pixels
- **Impulse noise:** contains random occurrences of white pixels
- **Gaussian noise**: variations in intensity drawn from a Gaussian normal distribution



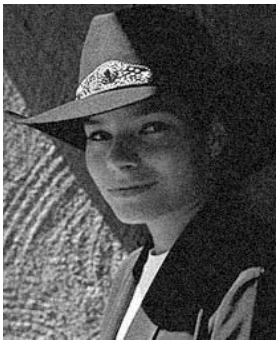
$f(x,y) = \overbrace{f(x,y)}^{\text{Ideal Image}} + \overbrace{\eta(x,y)}^{\text{Noise process}}$

Gaussian i.i.d. ("white") noise: $\eta(x,y) \sim \mathcal{N}(\mu, \sigma)$

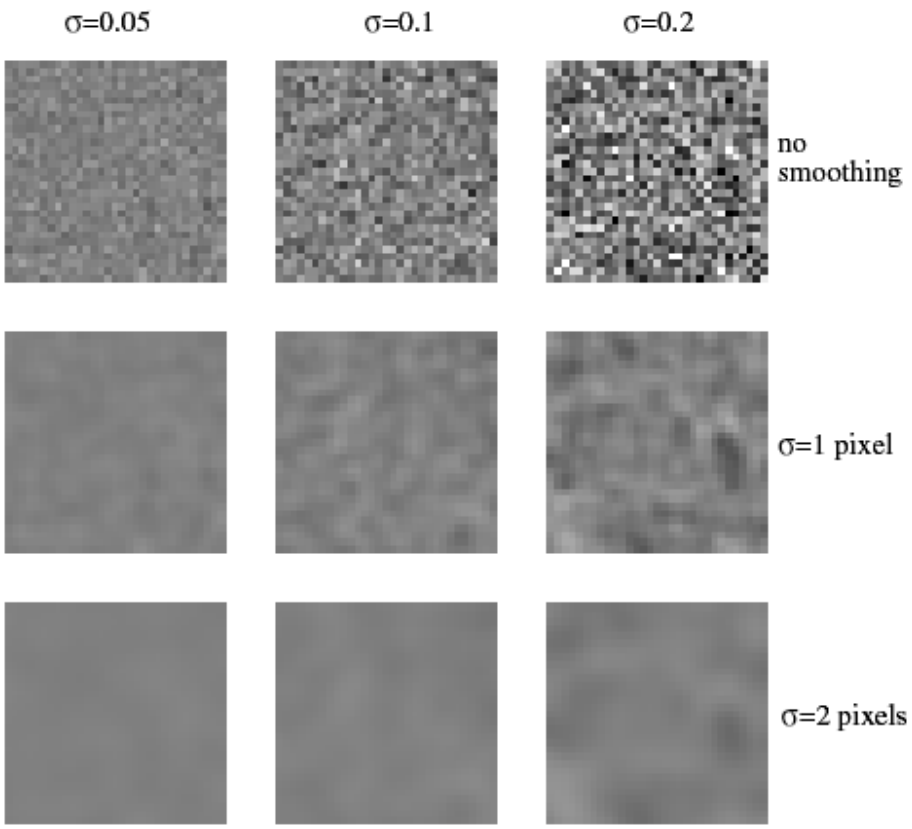

Original


Salt and pepper noise


Impulse noise


Gaussian noise

Smoothing (Blurring) : Gaussian Filter



Smoothing (Blurring) : Gaussian Filter

- **Salt and pepper noise**
- What's wrong with the results?



3x3 5x5 7x7



Median Filter

- Non-linear filter
- Efficient to remove salt and pepper noise

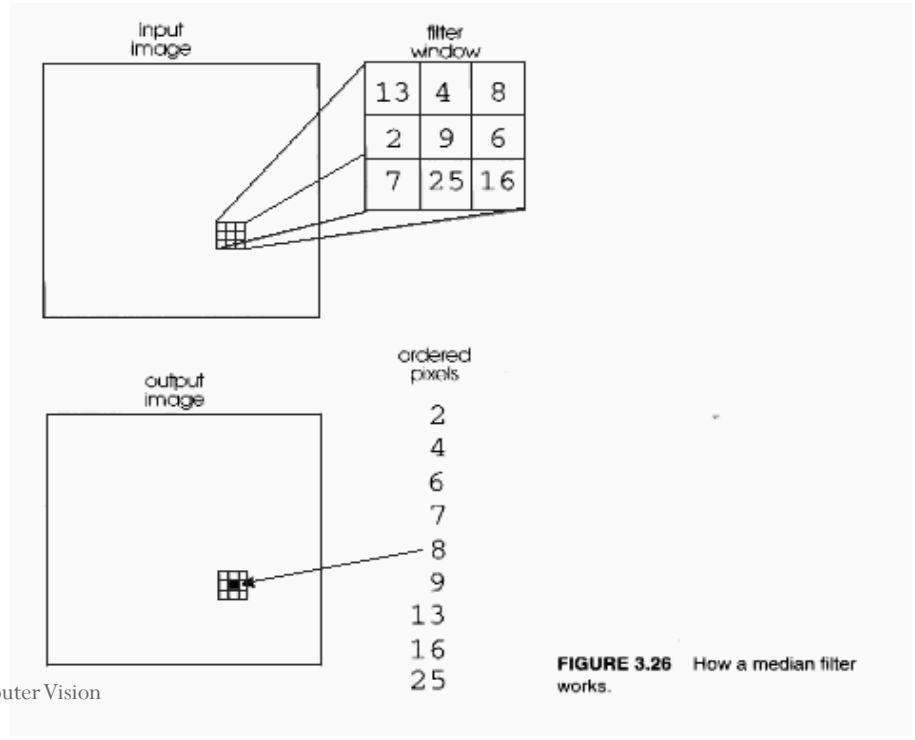


FIGURE 3.26 How a median filter works.

Median Filter

- Window of Median Filter

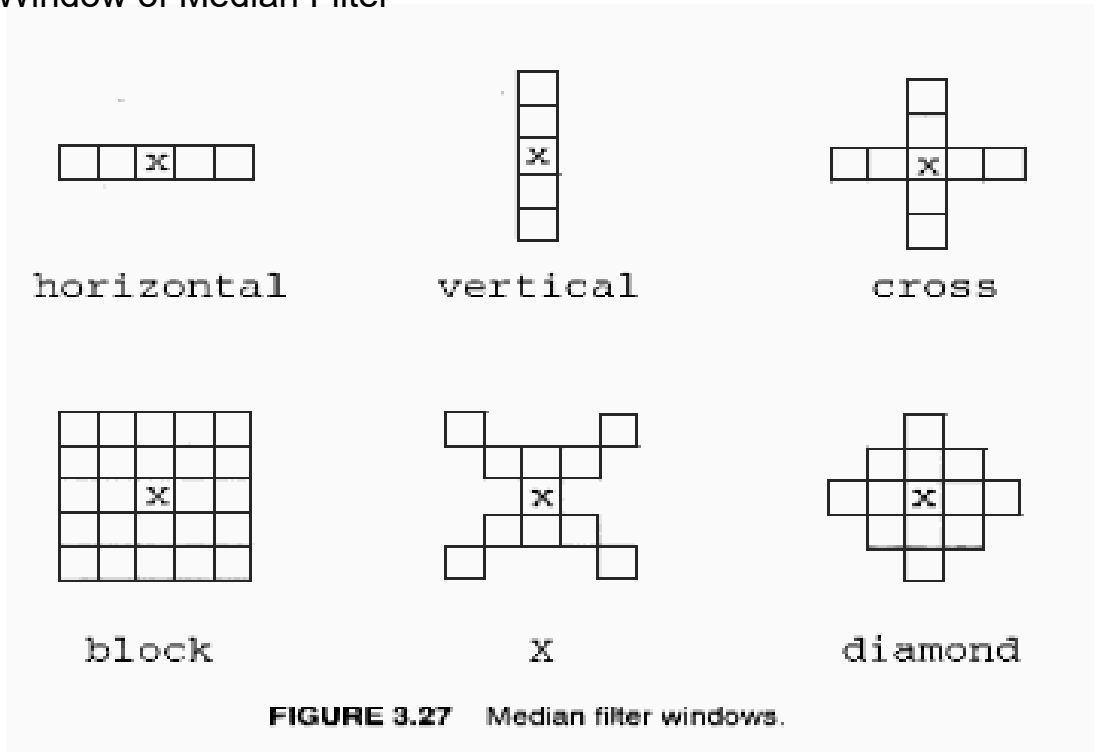
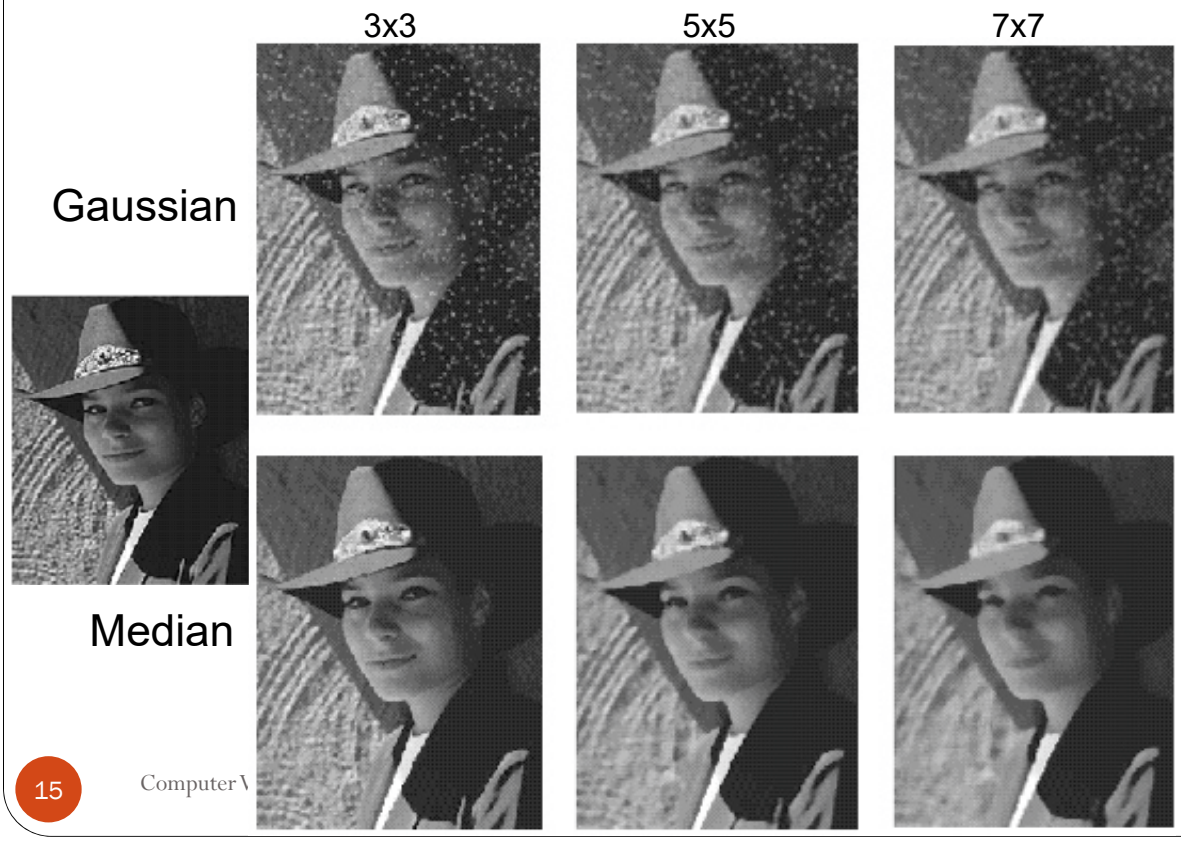


FIGURE 3.27 Median filter windows.

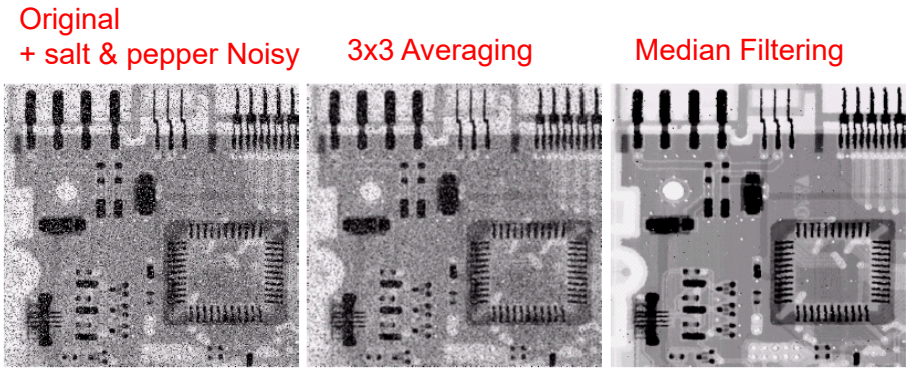
Gaussain Low-pass filter vs. Median Filter

- **Salt and pepper noise**



Averaging filter vs. Median Filter

- **Salt and pepper noise**



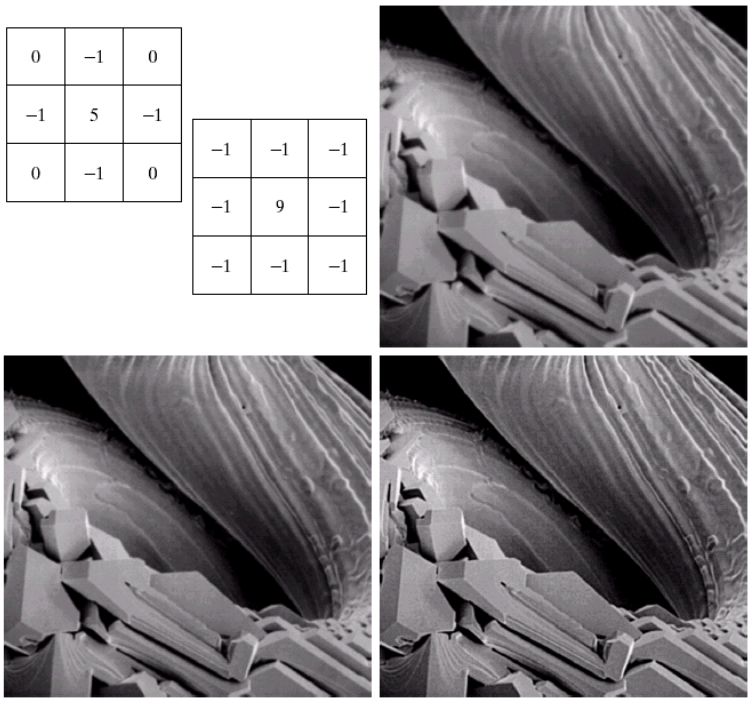
a b c

FIGURE 3.37 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3 × 3 averaging mask. (c) Noise reduction with a 3 × 3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Sharpening : High-pass Filter

0	−1	0
−1	5	−1
0	−1	0

−1	−1	−1
−1	9	−1
−1	−1	−1



a b c
d e

FIGURE 3.41 (a) Composite Laplacian mask. (b) A second composite mask. (c) Scanning electron microscope image. (d) and (e) Results of filtering with the masks in (a) and (b), respectively. Note how much sharper (e) is than (d). (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

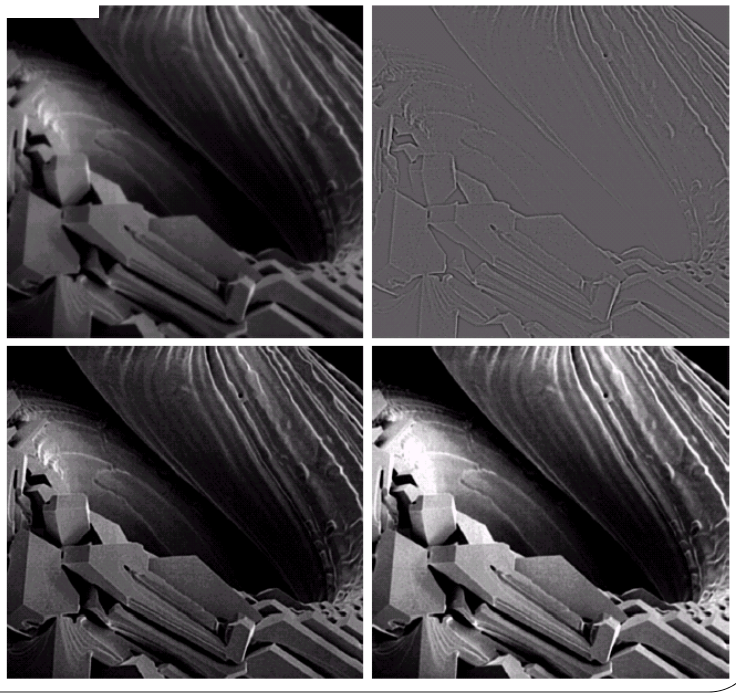
Sharpening : High-pass Filter

0	−1	0
−1	$A + 4$	−1
0	−1	0

−1	−1	−1
−1	$A + 8$	−1
−1	−1	−1

a b

FIGURE 3.42 The high-boost filtering technique can be implemented with either one of these masks, with $A \geq 1$.



c d

FIGURE 3.43
(a) Same as Fig. 3.41(c), but darker.
(a) Laplacian of (a) computed with the mask in Fig. 3.42(b) using $A = 0$.
(c) Laplacian enhanced image using the mask in Fig. 3.42(b) with $A = 1$. (d) Same as (c), but using $A = 1.7$.

Frequency Domain Filtering

- ✓ Fourier Transform
- ✓ Low-pass Filtering
- ✓ High-pass Filtering
- ✓ Band-pass Filtering
- ✓ Band-rejection Filtering
- ✓ Notch Filtering

이후 chapter 에서 배울 예정

Thank You
Next : Edge Detection



a b
c d

FIGURE 10.10
(a) Original image. (b) $|G_x|$, component of the gradient in the x-direction. (c) $|G_y|$, component in the y-direction. (d) Gradient image, $|G_x| + |G_y|$.

Proof: Fourier Transform of Gaussian Filter

ANNEX

$$G_\sigma = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

$$\int_{-\infty}^{\infty} e^{-u^2} du = ?$$

$$If \quad I = \int_{-\infty}^{\infty} e^{-u^2} du, \ then \quad I^2 = \int_{-\infty}^{\infty} e^{-s^2} ds \cdot \int_{-\infty}^{\infty} e^{-t^2} dt$$

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(s^2+t^2)} ds \cdot dt \qquad \because s = r\cos\theta, \ t = r\sin\theta$$

$$= \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r dr \cdot d\theta$$

Jacobian absolute

$$= 2\pi \int_0^{\infty} e^{-r^2} r dr \qquad \because \int f'(x) e^{f(x)} dx = e^{f(x)}$$

$$= \frac{2\pi}{-2} \int_0^{\infty} e^{-r^2} \cdot (-2r) dr$$

$$= -\pi \cdot e^{-r^2} \Big| = \pi$$

$$\therefore I = \sqrt{\pi}$$

Proof: Jacobian absolute value : Change of variables in Double Integral

ANNEX

$$\iint_R f(x,y) dx dy = \iint_{R^*} f(x(u,v), y(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv$$

$$J = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$\because x = r\cos\theta, \ y = r\sin\theta$$

$$\iint_R f(x,y) dx dy = \iint_{R^*} f(r\cos\theta, r\sin\theta) \left| \frac{\partial(x,y)}{\partial(r,\theta)} \right| dr d\theta \iint_{R^*} f(r\cos\theta, r\sin\theta) r dr d\theta$$

$$J = \left| \frac{\partial(x,y)}{\partial(r,\theta)} \right| = \begin{vmatrix} \frac{\partial r\cos\theta}{\partial r} & \frac{\partial r\cos\theta}{\partial \theta} \\ \frac{\partial r\sin\theta}{\partial r} & \frac{\partial r\sin\theta}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

Proof: Fourier Transform of Gaussian Filter

ANNEX

$$g(t)=\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{t^2}{2\sigma^2}} \quad \overset{F.T.}{\Leftrightarrow} \quad G(\omega)=e^{-\frac{\omega^2\sigma^2}{2}}$$

$$\Im(g(t))=\int_{-\infty}^{\infty}g(t)e^{-j\omega t}dt$$

$$=\frac{1}{\sqrt{2\pi}\sigma}\boxed{\int_{-\infty}^{\infty}e^{-\frac{t^2}{2\sigma^2}}e^{-j\omega t}dt}\ \Delta$$

$$=e^{-\frac{\omega^2\sigma^2}{2}}$$

$$u=\frac{t}{\sqrt{2}\sigma}+j2\sqrt{2}\pi f\sigma$$

$$\Downarrow$$

$$\frac{du}{dt}=\frac{1}{\sqrt{2}\sigma}\therefore dt=\sqrt{2}\sigma du$$

$$\Delta=\int_{-\infty}^{\infty}e^{-\frac{t^2}{2\sigma^2}}e^{-j\omega t}\,dt$$

$$=\int_{-\infty}^{\infty}e^{-\frac{t^2}{2\sigma^2}-j2\pi ft}\,dt$$

$$=\int_{-\infty}^{\infty}e^{-\left(\left(\frac{t}{\sqrt{2}\sigma}\right)^2+j2\pi ft+\left(j\sqrt{2}\pi f\sigma\right)^2\right)}\,dt$$

$$\cdot e^{\left(j\sqrt{2}\pi f\sigma\right)^2}$$

$$=\int_{-\infty}^{\infty}e^{-\left(\frac{t}{\sqrt{2}\sigma}+j\sqrt{2}\pi f\sigma\right)^2}\,dt\cdot e^{\left(j\sqrt{2}\pi f\sigma\right)^2}$$

$$=\int_{-\infty}^{\infty}\left(e^{-u^2}\cdot\sqrt{2}\sigma\right)du\cdot e^{\left(j\sqrt{2}\pi f\sigma\right)^2}$$

$$=\sqrt{2}\sigma\cdot e^{-2\pi^2\sigma^2f^2}\cdot\int_{-\infty}^{\infty}e^{-u^2}$$

$$\int_{-\infty}^{\infty}e^{-u^2}du=\sqrt{\pi}$$

$$=\sqrt{2\pi}\sigma\cdot e^{-\frac{\omega^2\sigma^2}{2}}$$

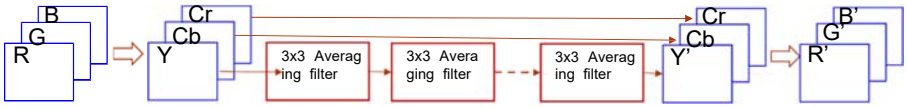
Computer Vision
Report 06: Filtering Smoothing

Yeungnam University
Dept. of Information and Communication
Ho-Youl JUNG

영남대학교 Yeungnam University

주요내용

- RGB (원본)➔ YCbCr ➔ Y 정보만으로 Averaging filtering ➔YCbCr ➔ RGB 영상 (복원)
- RGB (원본) 과 RGB 컬러 영상(복원) 의 **PSNR 구하는 프로그램** 작성
- 방법 : Y 영상을 입력 영상으로 하여 아래 그림과 같이 3x3 averaging filter를 적용



Report#6

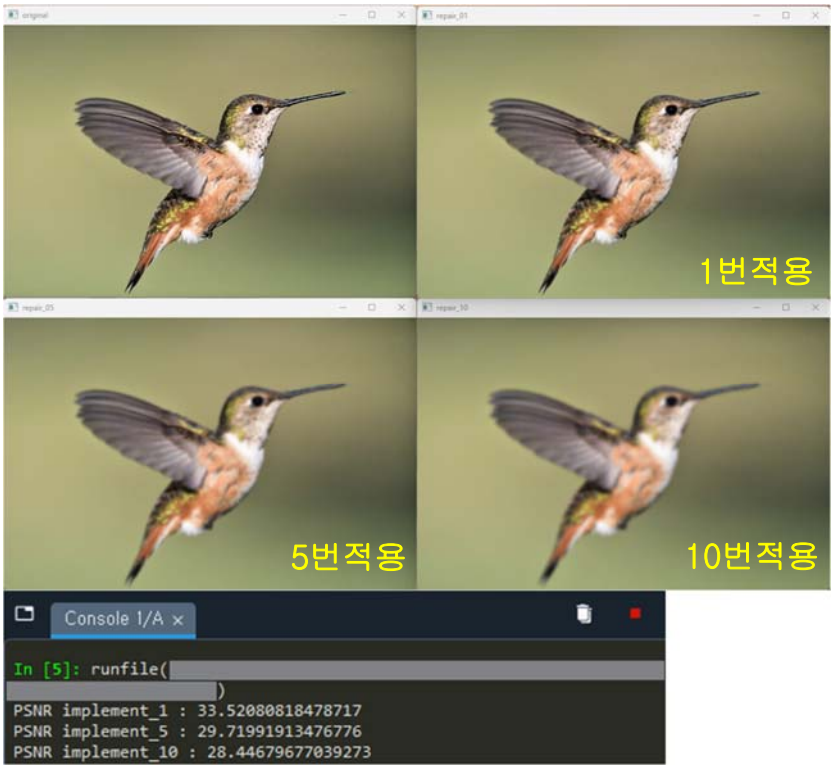
- 구현한 프로그램의 코드와 실행 결과를 캡처하고 워드 또는 아래 한글에 붙여 넣어 파일을 강의 지원에 제출
- 파일명 : report06_성명_학번.hwp

◆ 구현1: 3x3 averaging filter 을 1번 5번 10번을 적용하고 그 결과를 비교

◆ 구현2: 1번 5번 10번 적용한 이미지의 PSNR 비교

◆ Python 또는 C++ 중 하나만 구현

Report#6 실행결과 예)




```
Console 1/A ×
In [5]: runfile(
)
PSNR implement_1 : 33.52080818478717
PSNR implement_5 : 29.71991913476776
PSNR implement_10 : 28.44679677039273
```